Neo-81 — Full Hardware Validation Checklist

Pre-flight (do before any section)

- [ ] Latest firmware builds with 0 errors
- [ ] All host tests pass (`cmake -S App/Tests -B build-tests && cmake --build build-tests && ctest --test-dir build-tests`)
- [ ] Flash to board; power-cycle (USB unplug/replug — not just SWD reset)

Section 1 — Cursor rendering (P28)

Verifies `cursor_render()` refactor produces zero visible behaviour change across all 7 editors.

Main expression cursor

#	Test / Expected	✓
1	Power on, wait for home screen → Cursor blinks grey block on expression row	
2	Press `2ND` → Cursor turns amber with `^` inside	
3	Press any function key (e.g. `SIN`) → Cursor resets to grey block	
4	Press `ALPHA` → Cursor turns green with `A` inside	
5	Press any letter key (e.g. `A`) → Letter inserted, cursor resets to grey	
6	Press `STO→` → Cursor turns green with `A` inside (STO-pending state)	
7	While STO-pending, press any letter → Stores to that variable, cursor resets to grey block	
8	Press `INS` → Cursor changes to underscore style	
9	Press `INS` again → Cursor returns to full-height block	

Y= editor

#	Test / Expected	✓
10	Press `Y=` → Cursor blinks on first equation field	
11	Press `2ND` → Cursor turns amber with `^`	
12	Press any key to resolve 2ND → Cursor resets	
13	Press `INS` → Cursor changes to underscore style	
14	Press `INS` again → Cursor returns to block	

RANGE editor

#	Test / Expected	✓
15	Press `RANGE` → Cursor blinks in Xmin field	
16	Press `INS` → Cursor changes to underscore style	
17	Press `INS` again → Cursor returns to block	

ZOOM FACTORS editor

#	Test / Expected	✓
18	`ZOOM` → Select Factors → Cursor blinks in XFact field	
19	Press `INS` → Cursor changes to underscore style	
20	Press `INS` again → Cursor returns to block	

Matrix editor — insert mode must NOT change cursor shape

#	Test / Expected	✓
21	`MATRX` → EDIT → select matrix → navigate to a cell → Cursor is full-height grey block	
22	Press `INS` → Cursor shape does **not** change	
23	Press `2ND` → Cursor turns amber with `^`	
24	Press `ALPHA` → Cursor turns green with `A`	

PRGM name entry — insert mode must NOT change cursor shape

#	Test / Expected	✓
25	`PRGM` → EDIT → empty slot → ENTER → Cursor blinks in name field; full-height block	
26	Press `INS` → Cursor shape does **not** change — name entry does not support insert mode; the key…	

PRGM line editor

#	Test / Expected	✓
27	`PRGM` → EDIT → select a program → Cursor blinks on program line	
28	Press `INS` → Cursor changes to underscore style	
29	Press `INS` again → Cursor returns to block	

Section 2 — Parametric graphing (P35h)

#	Test / Expected	✓
1	`MODE` → row 4 → select `Param` → press `Y=` → Y= screen shows $X_1t=$ / $Y_1t=$ layout (not $Y_1=$)	
2	In param Y=, enter `cos(T)` for X_1t and `sin(T)` for Y_1t using `T` key → Equation accepted; T tok…	
3	Set `RANGE`: Tmin=0, Tmax=6.28, Tstep=0.13, Xmin=-2, Xmax=2, Ymin=-2, Ymax=2 → press `GRAPH` → Ci…	
4	Press `RANGE` while in param mode → Exactly 9 fields shown: Tmin, Tmax, Tstep, Xmin, Xmax, Xscl, …	
5	Press `TRACE` and use LEFT/RIGHT → Readout shows `T=` / `X=` / `Y=` (not just `X=/Y=`)	
6	`2nd+ON` to save → power-cycle → enter param Y= → X_1t/Y_1t equations intact	

Section 3 — DRAW menu (P29h)

<table>
<tr><th>#</th><th>Test / Expected</th><th>✓</th></tr>
<tr><td>1</td><td>Press <code>2ND+PRGM</code> → DRAW menu opens with 7 items: <code>ClrDraw</code>, <code>Line(</code>, <code>PT-On(</code>, <code>PT-Off(</code>, <code>PT-Chg… | | | 2 | Type</code>ClrDraw<code>from expression buffer (via DRAW menu → item 1) → ENTER → Draw layer cleared; resu… | | | 3 | Type</code>Line(0,0,5,5)<code>→ ENTER → Diagonal line drawn on graph canvas from (0,0) to (5,5) | | | 4 | Type</code>PT-On(2,3)<code>→ ENTER → Single pixel set at graph coordinates (2,3) | | | 5 | Type</code>PT-Off(2,3)<code>→ ENTER → That pixel cleared | | | 6 | Type</code>PT-Chg(2,3)<code>→ ENTER twice → Pixel toggles: on after first ENTER, off after second | | | 7 | Type</code>DrawF sin(X)<code>→ ENTER → Sine curve drawn as white overlay on graph canvas | | | 8 | Type</code>Shade(-1,1)<code>→ ENTER → Horizontal band between y=−1 and y=1 shaded on graph canvas (constan… | | | 9 | After drawing, press</code>ZOOM<code>→</code>Standard<code>→ return to graph → Draw layer content still visible (pe… | | | 10 | Type</code>ClrDraw<code>→ ENTER → All drawn content cleared from graph canvas | | | 11 | Set window X[−5,5] Y[−5,5] → type</code>Shade(sin(X),cos(X),2,-3,3)<code>→ ENTER → Shaded region appears o… | | | 12 | Type</code>Shade(X+1,X^3-8*X)<code>→ ENTER → Shading appears in columns where X+1 < X³−8X; boundary curves… | | | 13 | Type</code>Shade(sin(X),cos(X),1,-1,1)<code>→ ENTER, then</code>Shade(sin(X),cos(X),8,-1,1)` → ENTER → Resoluti…</td><td></td></tr>
</table>

Interactive cursor-pick DRAW (from graph canvas)

#	Test / Expected	✓
14	From graph canvas, press `2ND+PRGM` → select `2:Line(` → DRAW menu closes; crosshair cursor appea…	
15	Move cursor with arrow keys to a start point → press ENTER → Start point recorded (no visible cha…	
16	Move cursor to a different position → press ENTER → Line segment drawn between start and end poin…	
17	From graph canvas, press `2ND+PRGM` → select `3:PT-On(` → move cursor → press ENTER → Single pixe…	
18	With PT-On(cursor active, press `GRAPH` → Cursor-pick mode exits; free-cursor returns	
19	From graph canvas, `2ND+PRGM` → `4:PT-Off(` → move to a pixel set in test 17 → ENTER → That pixel…	
20	From graph canvas, `2ND+PRGM` → `5:PT-Chg(` → ENTER → ENTER at same point → Pixel toggles on then…	

Section 4 — STAT menu (P30h)

#	Test / Expected	✓
1	Press `2ND+MATRX` → STAT menu opens with 3 tabs: CALC / DRAW / DATA	
2	DATA tab → Edit → enter 5 pairs: (1,3), (2,5), (3,7), (4,9), (5,11) → All 5 pairs accepted and vi…	
3	CALC → `1:1-Var` → Results: n=5, x̄=3, Sx≈1.5811 (accept ±0.001)	
4	CALC → `2:LinReg` → Results: a=2, b=1, r=1; variables A=2 and B=1 set in calc engine (verify with…	
5	DATA → `3:xSort` → x-values sorted ascending; y-values reordered to match	
6	DATA → `4:ySort` → y-values sorted ascending; x-values reordered to match	
7	Set RANGE: Xmin=0, Xmax=6, Ymin=0, Ymax=12 → DRAW → `2:Scatter` → Scatter plot of data points vis…	
8	DRAW → `3:xyLine` → Points connected by lines on graph canvas	
9	DRAW → `1:Hist` → Histogram bars visible on graph canvas	
10	`2nd+ON` to save → power-cycle → STAT → DATA → Edit → All 5 data pairs intact	

#	Test / Expected	✓
11	DATA → Edit → navigate to row 3 → press RIGHT to reach column 3 (row-select state) → `>` prefix a…	
12	With `>` cursor at row 3 → press `INS` → New row (0,0) inserted before the current row 3; remaini…	
13	With `>` cursor at a row → press `DEL` → That row is removed; remaining rows shift up	
14	From home screen, after running 1-Var with 5 pairs: type `{x}(1)` → ENTER → Result is the first x…	

Section 5 — VARS menu (P31h)

#	Test / Expected	✓
1	Press `VARS` → 5-tab menu opens: XY / Σ / LR / DIM / RNG. LEFT/RIGHT navigate tabs; UP/DOWN navig…	
2	RNG tab → Shows 10 items. Scroll indicators (↑↓) appear when scrolling past 7. Item `0:Tstep` sho…	
3	XY tab → select `2:x̄` → Correct mean value (3) inserted into expression	
4	XY tab → select `3:Sx` → Correct Sx value (≈1.5811) inserted	
5	DIM tab → select `1:Arow` → Current row count of matrix [A] inserted	
6	DIM tab → select `2:Acol` → Current col count of matrix [A] inserted	
7	Press `Y=`, move cursor to Y_1= equation field, press `VARS` → XY → `2:x̄` → Value inserted into Y…	
8	LR tab → select `1:a` → Regression coefficient a=2 inserted	
9	LR tab → select `4:RegEQ` → String `aX+b` inserted into expression	
10	DIM tab → scroll to item `7:Dim{x}` → select → Current stat list length (5, if 5 pairs were enter…	

Section 6 — Y-VARS menu (P32h)

#	Test / Expected	✓
1	Press `2ND+VARS` → 3-tab menu opens: Y / ON / OFF. LEFT/RIGHT navigate tabs; UP/DOWN navigate ite…	
2	Y tab → select 1:Y_1 → String Y_1 inserted into expression buffer	
3	Store `3→X`; type Y_1 (via Y-VARS) → ENTER → Result is 3 (evaluates Y_1=X at X=3)	
4	Press `Y=`; move cursor into Y_1 equation field; press `2ND+VARS` → Y → select 2:Y_2 → Y_2 inser…	
5	OFF tab → `1:All-Off` → press `Y=` → All equations show `–` (disabled); no `=` visible	
6	ON tab → 2:Y_1-On → press `Y=` → Y_1 shows `=` (enabled); Y_2–Y_4 remain `–`	
7	Y tab → press `1` directly (no UP/DOWN first) → Y_1 inserted immediately — digit shortcut works …	
8	In Func mode: Y tab shows exactly 4 items (Y_1–Y_4); in Param mode: Y tab shows 10 items (Y_1–Y_4 the…	
9	In Param mode: Y tab → press `6` (digit shortcut for item 6) → X_1t string inserted immediately	
10	In Param mode: Y tab → scroll down to item 8:Y_2t → ENTER → Y_2t inserted into expression buffer	
11	In Param mode: ON tab → 8 items visible: `1:All-On`, 2:Y_1-On, 3:Y_2-On, 4:Y_3-On, 5:Y_4-On, …	
12	In Param mode: OFF tab → select 6:X_1t-Off → press `Y=` → X_1t/Y_1t pair shows `–` (disabled); oth…	

Section 7 — PRGM system (P10)

Menu & Navigation

#	Test / Expected	✓
T01	**Open PRGM menu**: From home, press `PRGM` → PRGM menu opens: `EXEC  EDIT  ERASE` tab bar, EXEC …	
T02	**Tab navigation with wrap**: With PRGM open, press RIGHT twice → Tab advances EXEC→EDIT→ERASE. L…	
T03	**Scroll indicators**: EXEC or EDIT tab → press DOWN past slot 7 → `↓` amber indicator at bottom …	
T04	**Close menu**: With PRGM open, press CLEAR → PRGM menu closes; home screen returns; expression u…	
T05	**Open PRGM from MATH menu**: MATH key → then press PRGM → MATH closes; PRGM opens on EXEC tab; n…	

Name Entry & Program Creation

#	Test / Expected	✓
T06	**Create new program with name**: PRGM → EDIT → empty slot → ENTER → type `TEST` → ENTER → Name-e…	
T06a	**Name-entry LEFT/RIGHT cursor**: On name-entry with `ABCD` typed, press LEFT twice → type `X` → …	
T07	**Create program — skip name**: EDIT → empty slot → ENTER → immediately ENTER again → Editor open…	
T08	**Digits allowed in name**: EDIT → empty slot → ENTER → type `A1B2` (ALPHA for letters, plain for…	
T08a	**DOWN from name-entry opens body**: Name-entry with `MYTEST` typed → press DOWN → Editor body op…	
T09	**Name-entry DEL**: Type a few letters → press DEL → Last character removed on first DEL press. A…	
T09d	**ALPHA_LOCK in name entry**: From name-entry, press `2ND+ALPHA` to engage ALPHA_LOCK → Cursor gr…	
T09b	**ENTER works on first press**: Type name → press ENTER exactly once → Editor opens on first ENTE…	
T09c	**CLEAR works on first press**: Type letters in name entry → press CLEAR → Returns to PRGM EDIT t…	
T10	**Cancel name entry (empty)**: From name-entry (no letters typed) → CLEAR → Returns to PRGM EDIT tab	
T11	**Open existing program**: EDIT tab → navigate to named slot → ENTER → Editor opens directly (no …	
T11b	**EDIT tab digit shortcut**: PRGM → EDIT tab → press `1` without UP/DOWN → Editor for slot 1 open…	

Editor

<table>
<tr><th>#</th><th>Test / Expected</th><th>✓</th></tr>
<tr><td>T12</td><td>Character input: Open any program → type `A+1` → Line shows `:A+1`; cursor advances</td><td></td></tr>
<tr><td>T12b</td><td>Insert mode off by default: Open program → navigate LEFT → type a character (no INS pressed) …</td><td></td></tr>
<tr><td>T13</td><td>CTL menu inserts keyword: In editor → PRGM → `3:If` (or press `3`) → CTL closes; line shows `… | | | T14 | **I/O menu inserts keyword**: In editor → PRGM → LEFT/RIGHT to I/O tab →`1:Disp`(or press`1`) … | | | T15 | **Multi-line scroll**: In editor → press ENTER 6+ times →`↓`indicator on last visible row;`↑`… | | | T16 | **Erase a program**: PRGM → ERASE tab → navigate to named slot → ENTER → Confirmation dialog:` 1:…</td><td></td></tr>
<tr><td>T16b</td><td>ERASE shows all 37 slots: Open PRGM → ERASE tab → All 37 slots visible (named and unnamed). N…</td><td></td></tr>
</table>

Persistence

#	Test / Expected	✓
T17	**Programs survive power cycle**: Create program `SAVE` in slot 1 with `Disp "OK"` → `2nd+ON` → U…	

Executor — Basic Execution

#	Test / Expected	✓
T18	**Run Disp program**: Slot 1 has `HELLO` with body `Disp "HELLO"` → EXEC → slot 1 → ENTER → ENTER…	
T19	**Run empty slot**: EXEC → slot with no body → ENTER → ENTER → `Done` appears; no error or lockup	
T20	**Expression evaluation and ANS**: Program: `2+2` then `Disp ANS` → run → `4` in history output; …	
T21	**Input and variable store**: Program: `Input A` then `Disp A` → run → type `7` → ENTER → `?` pro…	
T22	**If single-line skip**: Program: `0→A` / `If A=1` / `Disp "YES"` / `Disp "DONE"` → run → `YES` d…	
T23	**EXEC number-key shortcut**: PRGM → EXEC tab → press `1` (no UP/DOWN) → `prgm1` (or `prgmNAME`) …	
T24	**CLEAR aborts execution**: Program: `Lbl A` / `Disp "X"` / `Goto A` → run → press CLEAR while ru…	

Executor — Advanced Control Flow

#	Test / Expected	✓
T25	**Lbl/Goto label entry**: In editor → CTL → `Lbl` → type `A` → try to type a second letter → _(No…	
T26	**Disp string — left alignment**: Program: `Disp "HELLO"` → run → `HELLO` left-aligned, grey	

#	Test / Expected	✓
T27	**Disp variable — right alignment**: Program: `5→A` / `Disp A` → run → `5` right-aligned, white	
T28	**Goto/Lbl jump**: Program: `Goto END` / `Disp "SKIP"` / `Lbl END` / `Disp "DONE"` → run → `SKIP`…	
T29	**Pause halts and resumes**: Program: `Disp "WAIT"` / `Pause` / `Disp "RESUMED"` → run → Halts af…	
T30	**Stop terminates early**: Program: `Disp "A"` / `Stop` / `Disp "B"` → run → `A` appears; `B` doe…	
T31	**IS>(increment and skip**: Program: `1→I` / `IS>(I,2)` / `Disp "SKP"` / `Disp I` → run → I=1→2;…	
T32	**DS<(decrement and skip**: Program: `3→I` / `DS<(I,3)` / `Disp "SKP"` / `Disp I` → run → I=3→2;…	
T33	**Subroutine auto-return**: Slot 2: `Disp "SUB"`. Slot 1: `Disp "MAIN"` / `prgm2` / `Disp "BACK"`…	
T33b	**EXEC sub-menu tab**: In editor → PRGM → RIGHT → RIGHT → EXEC tab (third tab) highlighted. Shows…	
T34	**Nested subroutine (2 deep)**: Slot 3: `Disp "DEEP"`. Slot 2: `prgm3` / `Disp "MID"`. Slot 1: `p…	

Executor — I/O Commands

<table>
<tr><th>#</th><th>Test / Expected</th><th>✓</th></tr>
<tr><td>T35</td><td>prgmNAME execution (named): EXEC → named slot (e.g. <code>1:Prgm1 TEST</code>) → ENTER → ENTER → <code>prgmT… | | | T35b | **prgmNAME execution (unnamed)**: EXEC → unnamed slot (e.g.</code>3:Prgm3<code>) → ENTER → ENTER →</code>prgm3<code>… | | | T36 | **ClrHome clears history**: Program:</code>Disp “LINE1”<code>/</code>Disp “LINE2”<code>/</code>ClrHome<code>/</code>Disp “AFTER”<code>… | | | T37 | **DispGraph switches to graph view**: Ensure Y_1=X. Program:</code>DispGraph<code>/</code>Pause<code>/</code>DispHome` → …</td><td></td></tr>
</table>

Executor — End command

<table>
<tr><th>#</th><th>Test / Expected</th><th>✓</th></tr>
<tr><td>T38</td><td>End terminates at top level: Program: <code>Disp "A"</code> / <code>End</code> / <code>Disp "B"</code> → run → <code>A</code> appears; <code>B… | | | T39 | **End returns from subroutine**: Slot 2:</code>Disp “SUB”<code>/</code>End<code>. Slot 1:</code>Disp “MAIN”<code>/</code>prgm2` / …</td><td></td></tr>
</table>

PRGM MODE sub-menu

<table>
<tr><th>#</th><th>Test / Expected</th><th>✓</th></tr>
<tr><td>T40</td><td>PRGM MODE opens NUMBER tab: In editor → press <code>MODE</code> key → PRGM MODE sub-menu opens with <code>NUM… | | | T41 | **PRGM MODE NUMBER item inserts keyword**: Select</code>2:Sci<code>→</code>:Sci<code>inserted into current program … | | | T42 | **PRGM MODE GRAPH tab**: From NUMBER tab, press RIGHT → GRAPH tab active; 10 items:</code>1:Function<code>–… | | | T43 | **PRGM MODE executes notation change**: Slot 1:</code>Sci<code>. Run slot 1 from home. Type</code>123` → ENTER →…</td><td></td></tr>
</table>

Editor Alpha Behaviour

#	Test / Expected	✓
T41	**Cursor blinks in editor**: Open any program; observe for ~2 seconds → Cursor blinks at ~530 ms …	
T42	**2ND mode in editor**: In editor → press `2ND` → Cursor turns amber with `^`; resolves on next n…	
T43	**ALPHA single in editor**: In editor → press `ALPHA` once → Cursor green `A`; one letter inserte…	
T43b	**ALPHA_LOCK in editor**: In editor → press `2ND+ALPHA` → Cursor green `A`; multiple letters inse…	
T44	**Cursor blinks in name entry**: Open name-entry screen; observe for ~2 seconds → Cursor blinks a…	

Editor Body Behaviour

#	Test / Expected	✓
T45	**Body-only slot opens directly**: Editor → type content → CLEAR at name-entry (skip name). Reope…	
T46	**Tab wrap in sub-menus**: In editor → PRGM → CTL tab. Press LEFT. → Wraps to EXEC. LEFT→I/O; LEF…	
T46b	**EXEC digit and ALPHA+letter shortcuts**: On EXEC sub-menu → press `2` → `:prgm2` inserted immed…	

Executor — Input no-arg graph exploration (guidebook p. 8-13)

#	Test / Expected	✓
T47	**Input no-arg shows graph**: Program: `Input` (no variable) / `Disp X` / `Disp Y` → run → Graph …	

#	Test / Expected	✓
T48	**Cursor movement stores X and Y**: While in Input graph mode, move cursor a few pixels right and…	
T49	**CLEAR aborts program**: Program: `Input` → run → press CLEAR while graph cursor is active → Exe…	
T50	**Input no-arg mid-program**: Program: `Disp "START"` / `Input` / `Disp X` / `Disp "END"` → run →…	
T51	**X and Y variables updated in calc engine**: After T48, return to home screen → type `X` → ENTER…	

Section 8 — MODE screen — display and graph modes

Validates MODE rows 0, 5, 6, 8 and the free-roaming graph cursor introduced with MODE_GRAPH_FREE_CURSOR.

Display notation (Sci / Eng / Norm)

#	Test / Expected	✓
1	Press `MODE` → inspect row 0 → Three options shown: `Norm  Sci  Eng`; current selection highlighted	
2	Select `Sci` → ENTER → return to home → type `12345` → ENTER → Result shows `1.2345E4` (exactly o…	
3	In Sci mode, type `0.00123` → ENTER → Result shows `1.23E-3`	
4	Select `Eng` → ENTER → type `12345` → ENTER → Result shows `12.345E3` (exponent is a multiple of 3)	
5	In Eng mode, type `0.00123` → ENTER → Result shows `1.23E-3`	
6	Select `Norm` → ENTER → type `0.0005` → ENTER → Result shows `5E-4` (auto-sci threshold: values <…	
7	In Norm mode, type `0.001` → ENTER → Result shows `0.001` (threshold is strictly less than 0.001,…	
8	Set `Sci` → `2nd+ON` → power-cycle → check home screen with `100` → ENTER → Result shows `1E2` — …	

Free-roaming graph cursor

#	Test / Expected	✓
9	Press `GRAPH` from home screen → Blinking white crosshair appears at screen centre; `X=` and `Y=`…	
10	Press LEFT, RIGHT, UP, DOWN → Crosshair moves one pixel per press; `X=` and `Y=` update with each…	
11	From free-cursor position, press `TRACE` → Cursor snaps to nearest active equation at the current…	
12	While in TRACE mode, press `TRACE` again → Cursor re-snaps to X=x_mid on the same equation — does…	
13	While in TRACE mode, press `GRAPH` → Graph re-renders; blinking white free cursor appears at scre…	
14	While in TRACE mode, press a digit key (e.g. `5`) → Trace exits; `5` appears in expression buffer…	
15	From `Y=` screen, press `GRAPH` → Blinking free cursor appears (not a static render)	
16	From `RANGE` screen, press `GRAPH` → Blinking free cursor appears	
17	From `ZOOM` menu, press `GRAPH` → Blinking free cursor appears	
18	Navigate to home screen or any menu after graphing → Cursor blink timer stops — no blinking anima…	

Connected / Dot plot mode (P33h)

#	Test / Expected	✓
19	Press `MODE` → row 5 shows `Connected  Dot` → Both options visible; current selection highlighted	
20	Select `Dot` → ENTER → enter Y_1=sin(X) → set ZTrig preset → `GRAPH` → Only individual pixels plot…	
21	Select `Connected` → ENTER → re-`GRAPH` (or press GRAPH again) → Interpolated line segments conne…	
22	While in Dot mode, select MODE row 4 `Param` → ENTER → Y= layout switches to X_1t/Y_1t (row index b…	
23	Set `Dot` → `2nd+ON` → power-cycle → check MODE row 5 → `Dot` remains selected	

Sequential / Simultaneous plot mode (P38h)

#	Test / Expected	✓
24	Press `MODE` → row 6 shows `Sequential  Simul` → Both options visible	
25	Select `Simul` → ENTER → `GRAPH` → Both Y_1 and Y_2 advance across the screen simultaneously — visi…	

#	Test / Expected	✓
26	Select `Sequential` → ENTER → `GRAPH` → Y_1=X plots completely first, then Y_2=X^2 plots	
27	Set `Simul` → `2nd+ON` → power-cycle → check MODE row 6 → `Simul` remains selected	

Polar coordinate display (P40h)

#	Test / Expected	✓
28	Press `MODE` → row 8 shows `Rect  Pol` → Both options visible	
29	Select `Pol` → ENTER → `GRAPH` with Y_1=X^2 → move free cursor → Readout below canvas shows `R=` an…	
30	In Pol display mode, press `TRACE` and move LEFT/RIGHT → Trace readout shows `R=` / `θ=` for each…	
31	Select `Rect` → ENTER → `GRAPH` → Readout reverts to `X=` / `Y=`	
32	Set `Pol` → `2nd+ON` → power-cycle → check MODE row 8 → `Pol` remains selected	
33	On home screen (Rad angle mode), type `R>P(−1,0)` → ENTER → Result is `1`; then type `θ` → ENTER:…	
34	Type `P>R(1,0)` → ENTER → Result is `1`; then type `Y` → ENTER: result is `0`	
35	Type `3` → `STO→` → `θ` → ENTER → type `θ` → ENTER → `θ` variable stores 3; second ENTER returns `3`	

TRACE auto-pan at viewport edge (guidebook p. 3-10)

#	Test / Expected	✓
36	In TRACE mode, hold RIGHT until cursor reaches Xmax → When cursor hits Xmax, viewport pans right …	
37	In TRACE mode after right-pan, hold LEFT until cursor reaches new Xmin → Viewport pans left by (X…	
38	Parametric mode: TRACE cursor at T boundary → Cursor clamps at Tmin/Tmax — no pan in parametric mode	

ZOOM cursor-pick (guidebook pp. 3-13, 3-14, 3-16)

#	Test / Expected	✓
39	Press `ZOOM` → `2:Zoom In` → Graph closes menu; blinking crosshair cursor appears on canvas (curs…	
40	Move cursor to a point of interest → press ENTER → Graph re-renders zoomed in; the picked point b…	
41	Press `ZOOM` → `3:Zoom Out` → move cursor → ENTER → Graph re-renders zoomed out centred on picked…	
42	Press `ZOOM` → `8:Integer` → move cursor → ENTER → Graph re-renders with integer pixel-to-coordin…	
43	While in ZOOM cursor-pick, press `GRAPH` (or any graph-nav key) → Cursor-pick mode exits without …	

Section 9 — MATH extensions (HYP tab + nDeriv + expression fixes)

MATH HYP tab

#	Test / Expected	✓
1	Press `MATH` → navigate to `HYP` tab → HYP tab shows 6 items: `sinh(`, `cosh(`, `tanh(`, `asinh(`…	
2	Select `1:sinh(` → type `0` → ENTER → Result is `0`	
3	Select `2:cosh(` → type `0` → ENTER → Result is `1`	
4	Select `3:tanh(` → type `0` → ENTER → Result is `0`	
5	Select `4:asinh(` → type `1` → ENTER → Result ≈ `0.881374`	
6	Select `5:acosh(` → type `1` → ENTER → Result is `0`	
7	Select `5:acosh(` → type `0` → ENTER → `DOMAIN ERR` (acosh domain is x ≥ 1)	
8	Select `6:atanh(` → type `0` → ENTER → Result is `0`	
9	Select `6:atanh(` → type `1` → ENTER → `DOMAIN ERR` (atanh domain is	
10	Type `sinh(1)` directly in expression buffer without menu → Result ≈ `1.1752` (confirms keyword t…	

nDeriv(

#	Test / Expected	✓
11	Type `nDeriv(X^2,X,3)` → ENTER → Result ≈ `6` (derivative of X^2 at X=3 is 2X=6; tolerance ±0.01)	
12	Type `nDeriv(sin(X),X,0)` → ENTER → Result ≈ `1` (derivative of sin at 0 is cos(0)=1; tolerance ±…	
13	Type `nDeriv(X^3,X,2)` → ENTER → Result ≈ `12` (derivative of X^3 at X=2 is $3X^2$=12; tolerance ±0.01)	

Auto-close trailing parenthesis

#	Test / Expected	✓
14	Type `sin(1` (no closing `)`) → ENTER → Result ≈ `0.841471` — unmatched `(` is auto-closed; no SY…	
15	Type `(1+2` → ENTER → Result is `3` — leading unmatched `(` auto-closed	
16	Type `sin(cos(0` → ENTER → Result ≈ `0.841471` — two unmatched `(` auto-closed in order	

Angle-override postfix tokens: `°` and `r` (guidebook pp. 2-3, 2-5)

#	Test / Expected	✓
17	Set angle mode to **Rad**. Type `sin(90°)` → ENTER → Result is `1` — `°` overrides RAD mode; calc…	
18	Set angle mode to **Deg**. Type `sin(1r)` → ENTER → Result ≈ `0.8415` — `r` overrides DEG mode; c…	
19	In Rad mode. Type `cos(180°)` → ENTER → Result is `-1` — 180° in radians is π; cos(π)=-1	
20	In Deg mode. Type `cos(3.14159r)` → ENTER → Result ≈ `-1` — π radians overridden to radians in DE…	

Section 10 — Matrix row operations (MATRX MATH menu)

#	Test / Expected	✓
1	Press `MATRX` → MATRX menu opens; navigate to MATH tab	
2	Select `1:det(` → type `[A])` → ENTER → Result is `-2` (det of [[1,2],[3,4]] = 1×4 − 2×3 = −2)	
3	Select `2:T` → type `[A])` → ENTER → Result is [[1,3],[2,4]] — transpose of [A]	
4	Select `3:rowSwap(` → type `[A],1,2)` → ENTER → Result is [[3,4],[1,2]] — rows 1 and 2 swapped	
5	Select `4:Row+(` → type `[A],1,2)` → ENTER → Result is [[1,2],[4,6]] — row 1 added to row 2	
6	Select `5:*Row(` → type `2,[A],1)` → ENTER → Result is [[2,4],[3,4]] — row 1 multiplied by scalar 2	
7	Select `6:*Row+(` → type `2,[A],1,2)` → ENTER → Result is [[1,2],[5,8]] — 2×row 1 added to row 2	
8	All row-op results: original [A] unchanged after evaluating expressions → Press `[A]` → ENTER: st…	

Section 11 — RESET menu

#	Test / Expected	✓
1	Press `2ND` then `+` from home screen → RESET MEMORY screen appears with title `MEMORY`, stat poi…	
2	On RESET screen: press `1` or `CLEAR` → RESET screen closes; returns to previous screen; no data …	
3	Stat count display → Open RESET screen when 5 stat pairs are entered	
4	Program count display → Open RESET screen when 2 programs are named	
5	Press `2` to confirm reset → RESET executes; home screen shows `Mem cleared`; ANS=0	
6	After reset: verify stat cleared → STAT → DATA → Edit	
7	After reset: verify variables cleared → Type `A` → ENTER	
8	After reset: verify Y= cleared → Press `Y=`	
9	After reset: verify RANGE defaults → Press `RANGE`	
10	`2ND++` accessible from non-home screens → Open STAT menu → press `2ND` then `+`	

Section 12 — Error overlay (guidebook pp. 1-26, B-4)

#	Test / Expected	✓
1	Type `sin(` (unclosed — but auto-close gives a valid result); instead type `1++1` → ENTER → Error…	
2	On error overlay, press `2` or `CLEAR` → Overlay closes; home screen returns; expression buffer u…	
3	Type `log(-1)` → ENTER → Error overlay shows `ERROR 02 DOMAIN`	
4	Type `1/0` → ENTER → Error overlay shows `ERROR 01 DIVIDE BY 0`	
5	Type `1++1` → ENTER → press `1` (Goto Error) → Overlay closes; cursor is positioned at the error …	
6	Trigger a DIMENSION error: with [A] set to 2×2, type `[A]+[B]` where [B] is 3×3 → ENTER → Error o…	
7	Trigger error during PRGM execution: program body `Disp 1/0` → run → Error overlay appears mid-ex…	

#	Test / Expected	✓
8	Trigger error during PRGM execution → press `1` (Goto Error) → Returns to home screen (program co…	

Section 13 — STO advanced targets: matrix, element, Y= (guidebook pp. 6-5, 6-10, 3-18)

#	Test / Expected	✓
1	Type `[[1,2],[3,4]]` → `STO→` → `[A]` (MATRX → NAMES → [A]) → ENTER → `[[1,2],[3,4]]` stored in […	
2	Type `5` → `STO→` → `[A]` → ENTER → All elements of [A] filled with `5` (scalar broadcast); press…	
3	Type `7` → `STO→` → `[A]` → `(` → `1` → `,` → `2` → `)` → ENTER → Element A stores 7; pres…	
4	Type `2+3` → `STO→` → `[A]` → `(` → `2` → `,` → `1` → `)` → ENTER → Expression evaluated to 5; [A…	
5	Type `X^2` → `STO→` → Y_1 (via `2ND+VARS` → Y tab → `1:`Y_1) → ENTER → Y_1 equation updated to `X^… | | | 6 | STO→ to out-of-bounds element: type` 1 → STO→ → [A] → (→ 9 → , → 1 →)` → ENTER → …	